A-A-51428
11 August 1986

COMMERCIAL ITEM DESCRIPTION

PORCELAIN ADJUSTMENT SET, DENTAL

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a porcelain adjustment set suitable for adjusting and polishing porcelain.

Salient characteristics:

Shall be a dental porcelain adjustment set containing three assorted, straight handpiece, aluminum oxide stones for making fine adjustments of porcelain; and nine assorted, straight handpiece, abrasive impregnated, silicone rubber points for polishing the adjusted porcelain areas.

One each of the assorted, straight handpiece, aluminum oxide stones shall be cone shaped; wheel shaped (rounded edge); and flame shaped.

Three each of the assorted, straight handpiece, points shall be bullet shaped; wheel shaped; and knife-edged. In groups of one each of the three shapes, the points shall be impregnated with medium grit, fine grit, and superfine grit. All points and stones shall be securely mounted on straight handpiece shanks and shall be suitable for use in wet or dry conditions at a maximum speed of 30,000 rpm. The stones shall be white in color and the points shall be light gray in color. The points shall be coded for grit size, such as, a double band of color on shank for superfine grit, a single band for fine grit, and a plain shank for medium grit. A code identification sheet shall be provided with each unit of issue. The stones and rubber points shall be mounted in a suitable plastic and plastic covered holder. Each unit of issue shall contain instructions for use. Diameter of the shanks shall be 2.334 to 2.350 mm (0.0919 to 0.0925 in). Other dimensions of the stones and points shall be as shown in table I.

Table I. Dimensions of stones and points.

Type	Shape	Overall Length	Head Diameter	Head Length
Stones	Cone	46.2 - 46.4 mm	2.3 ± 1.0 mm*	8.0 ± 0.1 mm
	Wheel	42.1 - 42.3 mm	6.2 ± 1.0 mm	2.1 ± 0.1 mm
	Flame	45.0 - 45.1 mm	3.0 ± 1.0 mm*	6.6 ± 0.1 mm
Points	Bullet	50.1 - 50.5 mm	5.0 ± 1.0 mm	16.3 ± 0.1 mm
	Wheel	42.5 - 42.7 mm	12.0 ± 2.0 mm	1.9 ± 0.1 mm
	Knife-edge	42.0 - 42.4 mm	12.6 ± 3.0 mm	3.5 ± 0.1 mm*

AMSC N/A FSC 6520

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*Nominal diameter at tips of cone and flame shapes, respectively, are 0.50 and 0.56 mm. Nominal thickness at edge of knife-edge is 0.50 mm.

Points shall be impregnated with the following grit sizes: (1) medium grit, 44 - 52 microns; (2) fine grit, 26-31 microns; and (3) superfine grit, 6 - 7 microns. The aluminum oxide grit for the stone shall be fine grit, 26-31 microns. The design of the porcelain adjustment set shall be as shown in figure 1.

Performance test. In accordance with manufacturer's instructions, points and stones shall produce a final glaze-like finish on porcelain. While in use, points and stones shall be examined for concentricity.

Runout. The total indicated runout of the stone shall not exceed 0.08 mm (0.003 in.) when held in a V block or precision chuck. The measurement shall be made at the base of the stone with a suitable measuring device sensitive to 0.005 mm (0.0002 in).

Material. The shanks shall be fabricated of corrosion-resisting stainless steel. The stone shall be micrograined aluminum oxide grit in a porcelain binder. The points shall be fabricated of silicone rubber with impregnated grit suitable for the purpose intended.

Workmanship. Porcelain adjustment set shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Set (SE). One set, as specified, constitutes one unit.

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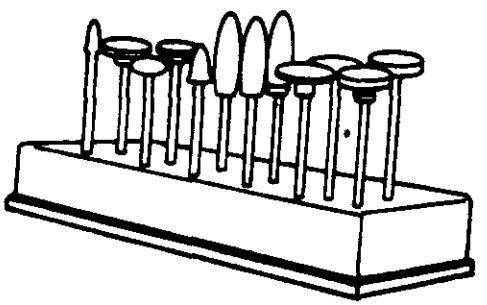


Figure 1. Porcelain Adjustment Set

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one porcelain adjustment set. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be

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responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packing, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

6520-01-119-7676	PORCELAIN ADJUSTMENT SET, DENTAL, Set of 12

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY CUSTODIANS:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITY

VA-OSS

Project No. 6520-1913

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